

# **FC4452 - SOFTWARE PROJECT MANAGEMENT**

## **QUESTION BANK**

### **UNIT I- SOFTWARE DEVELOPMENT PROCESSP (2 MARKS)**

1. What is the prime objective of software engineering?
2. Define software engineering paradigm.
3. What do you mean by spiral model?
4. Write a brief note on waterfall model.
5. Distinguish between process and methods.
6. Give the importance of software engineering.
7. Define software process. State the important features of a process.
8. Write any two characteristics of software as a product.
9. List the process maturity levels in SEI' s CMM.
10. Distinguish clearly between verification & validation.
11. What are the functions of data architecture?
12. Define System Modeling.
13. State the System Engineering Hierarchy.
14. Mention some of the factors to be considered during System Modeling.
15. What are the different architectures developed during BPE?
16. Define Verification &Validation.

### **4 MARKS**

1. Explain iterative waterfall and spiral model for software life cycle and discuss various activities in each phase. (16)
2. List several software process paradigms. Explain how both waterfall model and prototyping model can be accommodated in the spiral process model. (16)
3. Explain in detail Boehm's spiral model for softwre life cycle and discuss various activities in each phase. (16)
4. a) Which is more important-the product or process? Justify your answer. (4)  
b) Identify the umbrella activities in software engineering process. (4)  
c) With suitable illustration explain SPIRAL model evolutionary software development . (8)

### **UNIT II - SOFTWARE REQUIREMENT ENGINEERING(2MARKS)**

1. Mention any two non-functional requirements on software to be developed
2. What is known as SRS review? How is it conducted?
3. Distinguish between expected requirements and excited requirements
4. What is meant by software prototyping?
5. What are the non-functional requirements of software?
6. What is data dictionary? How is it used in software engineering?
7. Write the distinct steps in requirements engineering process?
8. Compare evolutionary and throw away prototyping?
9. What is the role of data dictionary?

10. Write a brief note on data modeling activity?
11. What is meant by Information flow Continuity?
12. Draw a DFD & CFD of a test monitoring system for Gas Turbine
13. Define Behavioral Modeling.
14. Draw the Context level DFD for the Safe home Software.
15. What is meant by Data dictionary?
16. Define Process Specification.
17. What does data dictionary contains?
18. Write down the Data dictionary for the data item "Telephone
19. What is meant by Throw away Prototyping?

#### **4 MARKS**

- 1) Explain the ways and means for collecting the software requirements and how are they organized and represented? (16)
- 2) Describe various prototyping techniques and discuss on analysis sand modeling.(16)
- 3) a) Discuss in detail the data modeling activity (8)  
b) Write briefly about the utility of state transition diagram in analysis modeling activity (8)
- 4) a) Compare functional and behavioral models. (4)  
b) With a suitable diagram explain the elements of the analysis model (8)  
c) With an example explain about DFD. (4)
- 5) a) Explain the extensions of DFD for real time systems. (8)  
b) Discuss the features of state transition diagram and its application. (8)
- 6) a) Explain the feasibility studies. What are the outcomes? Does it have either implicit or explicit effects on software requirement collection? (8)  
b) What is the prototyping technique? How prototype models are prepared for a software process? Discuss. (8)
- 7) a) Describe how software requirements are documented? State the importance of documentation. (8)  
b) Explain the software requirement analysis and modeling. (8)
- 8) Explain the state oriented approaches for representing behavioral specifications of software. (16)
- 9) Narrate the importance of software specification of requirements. Explain a typical SRS structure and its parts. (16)
- 10) Discuss in detail the FAST method of  
a. Requirement elicitation with an example. (8)  
b. What is software specification? (8)
- 11) Write short notes on data modeling? (6)  
a. Discuss in detail the basic structure of analysis model. (10)
- 12) How is SRS for a development project arrived at? (6)  
a. What minimum features are required to be present in a good SRS?(10)

#### **UNIT III- SOFTWARE MODELLING AND DESIGN (2 MARKS)**

- 1) What do you mean by horizontal and vertical partitioning?
- 2) How do you evaluate user interface?
- 3) Why software architecture is important in a software process?

- 4) Distinguish between horizontal and vertical partitioning?
- 5) How reliability is related to quality assurance?
- 6) Distinguish between horizontal and vertical partitioning?
- 7) What is the software architecture?
- 8) Compare data flow oriented design and data structured oriented design
- 9) What is the role of verification during a software exercise?
- 10) Distinguish between hard and soft real time systems.
- 11) Distinguish between product and process metrics.
- 12) What do you mean by reuse of design and update of a design as per technology changes and customer psychology?
- 13) What is the work product of software design process and who does this?
- 14) Enumerate different data flow architectures
- 15) How do you describe software interface?
- 16) How the requirements are collected for user interface of software?
- 17) What is transaction mapping? How it is used in software design?
- 18) What are the various models produce by the software design process?
- 19) What is the quality parameters considered for effective modular design?

#### **4 MARKS**

1. What is transform mapping? Explain the process with an illustration. What is its strength and weakness? (16)
2. a) Explain about the various design concepts considered during design? (12)  
b) Write short notes on user interface design process? (4)
3. a) Explain data architectural and procedural design for a software? (8)  
b) Describe the design procedure for data acquisition system (8)
4. Explain the importance of user interface design in sale of software. (16)
5. Describe decomposition levels of abstraction and modularity concepts in software design? (16)
6. What are the characteristics of a good design? Describe different types of coupling and cohesion. How design evaluation is performed? (16)
7. Draw the basic structure of analysis model and explain each entity in detail.(16)
8. a) discuss in detail about the design process in software development process (8)  
b) Justify "Design is not coding and coding is not design". (8)
9. a) Explain in detail about the characteristics and criteria for a good design. (10)  
b) Describe the golden rules for interface design. (6)
10. a) What is the design document?  
b) How is it organized?
11. What are the various software architectures available for the developer according to you? Which is the best and why? (8)
12. What do you mean by modularity in software development? Why is it needed? What is its strength? (8)
13. a) What are the various model of abstraction? Discuss any two in detail? (8)  
b) How does a real time system design differ comparing distributed system design? (8)
14. a) Explain the set of principles for software engineering design? (10)  
b) Describe the concept of information hiding. (6)
15. a) What is data flow oriented design. (3)

- b) Draw a detailed dataflow diagram for library management. (10)
- c) What are the components dataflow oriented design. (3)
- 16. a) What is SCM? (4)
- b) How are SCM tasks practiced over the operational life of software? (6)
- c) What is the impact of requirement changes during development of a software product? (6)

#### **UNIT IV- SOFTWARE PROJECT MANAGEMENT (2 MARKS)**

1. List out the importance of cost estimation in software development.
2. Mention the advantages of CASE tools.
3. How do you estimate time required for a software development project?
4. Draw the structure of CASE REPOSITORY and explain.
5. What is meant by software change?
6. Write short notes on empirical estimation models.
7. Why the software needs maintenance?
8. Define software re-engineering.
9. List any 4 categories of CASE tools.
10. What is Delphi cost estimation technique?
11. What is CASE?

#### **4 MARKS**

1. Explain the need for software measures and describe various metrics. (16)
2. Discuss briefly on software maintenance activities and how do you estimate the cost involved. (16)
3. a) Explain in detail about the maintenance process.(8)
- b) Discuss in detail about software evolution. (8)
4. Describe two metrics which are used to measure the software in detail. Discuss clearly the advantages and disadvantages of these metrics. (16)
5. a) What is Halsted's software science metric. Define. (8)
- b) Explain about function point metric in detail. (8)
6. Write short notes on
  - a) Software maintenance (8)
  - b) Task scheduling with an example. (8)
7. Explain various cost estimation models and compare. (16)
8. Write briefly on
  - a) CASE (8)
  - b) Software complexity measure. (8)
9. Explain the maintenance activities and maintenance problems. How the cost of maintenance is estimated? (16)
10. Write short notes on
  - a) COCOMO estimation criteria. (8)
  - b) Software metrics (8)
11. a) Justify the statement "Software maintenance is costlier". (8)
- a) Discuss the concept of software maintenance process. (8)

#### **UNIT V PROJECT SCHEDULING AND REENGINEERING (2 Marks)**

1. Define four P's of software project management
2. Define any two categories of Stakeholders
3. Enlist Different Characteristics of Team Leader
4. What is meant by software scope?
5. What is meant by problem Decomposition?
6. Define the term Process Decomposition
7. Define compartmentalization principle of project scheduling

8. What is meant by effort distribution in project scheduling?
9. Define two characteristics of Risk
10. What is Risk Identification?
11. Write any two steps to migrate risk
12. Define Reactive and Proactive Risk
13. Define the term Software Configuration Management
14. Define formal method

#### **4 Marks Question**

1. Explain the Term People in software project management
2. Describe different categories of stakeholders
3. What do we look for when choosing someone to team leader in software project?
4. What factors should be considered when the structure of a software team is chosen?
5. Explain the term software scope and problem decomposition
6. Explain process decomposition in software project management
7. What are reason that project is in Trouble?
8. Explain five part approach to avoid problems in software project
9. What are the reasons for late software delivery?
10. Explain basic principles of software project scheduling
11. How to define a Task set for software project?
12. Explain Task Network with example
13. Explain how to track scheduling of software project
14. Explain different types of risk encounter when we build software
15. Describe risk components
16. Explain the concept of risk projection
17. Explain Risk refinement
18. Explain Risk Mitigation and Monitoring
19. What is SCM ?explain elements of configuration management system
20. What functions are implemented by SCM Repository?

#### **UNIT VI SOFTWARE QUALITY ASSURANCE AND SECURITY (2 Marks)**

1. What is quality assurance?
2. State the different types of quality cost.
3. What is failure cost?
4. State the steps for software statistical quality assurance.
5. Define quality assurance system.
6. State the purpose of ISO 9000 quality standards.
7. Define software scope.
8. State four quality concepts.
9. State software quality.
10. What is feasibility?
11. State the activities of SQA(any four).
12. State decomposition techniques.
13. State the reusable software resources categories.

#### **4 Marks**

1. Explain basic quality concepts.
2. Explain the software quality assurance.
3. Explain the core steps of six sigma methodology.
4. Explain software safety.

5. What is software reliability?
6. Explain the project planning process.
7. Explain the software scope and feasibility.
8. Why we use decomposition technique? Explain any one in detail.
9. Explain problem based decomposition technique.
10. Explain process based decomposition technique.
11. Explain statistical software quality assurance with example.
12. .Explain any three decomposition techniques.
- 13.Explain project planning process with example.
- 14.State software realibility. Explain different software quality assurance activities.
- 15.Explain the following term
  - 1) Measures of reliability and Availability 2) Software Safety.